

AlgoMate

A Python command-line toolkit for number and list operations

Vikram Singh | 26BAI10973

Python Essentials | 1st Semester

Faculty: Rizwan Zaman Wani



Introduction

AlgoMate brings common arithmetic and list-processing tasks into one guided terminal program.

PROJECT AT A GLANCE

- Python 3 command-line application
- Two feature areas: Number Tools and List Tools
- Menu-driven input and immediate printed results
- Uses Python standard-library math functions and built-in list operations

Problem statement and objectives

PROBLEM

A learner who needs a quick calculation or basic list operation must otherwise switch between separate tools or write a short script for each task.

OBJECTIVES

- Combine common number utilities in one menu
- Provide practical list operations through a second menu
- Show inputs and computed outputs clearly
- Handle invalid numeric input and selected domain errors

Scope: an interactive local CLI; no database, network service, or graphical interface.

Functional requirements

NUMBER TOOLS

- GCD and LCM for two integers
- Prime check and prime factorization
- Fibonacci value and sequence
- Factorial, square root, and integer power

LIST TOOLS

- Reverse and sort a list
- Remove repeated values
- Find maximum and count occurrences
- Find the kth-smallest value

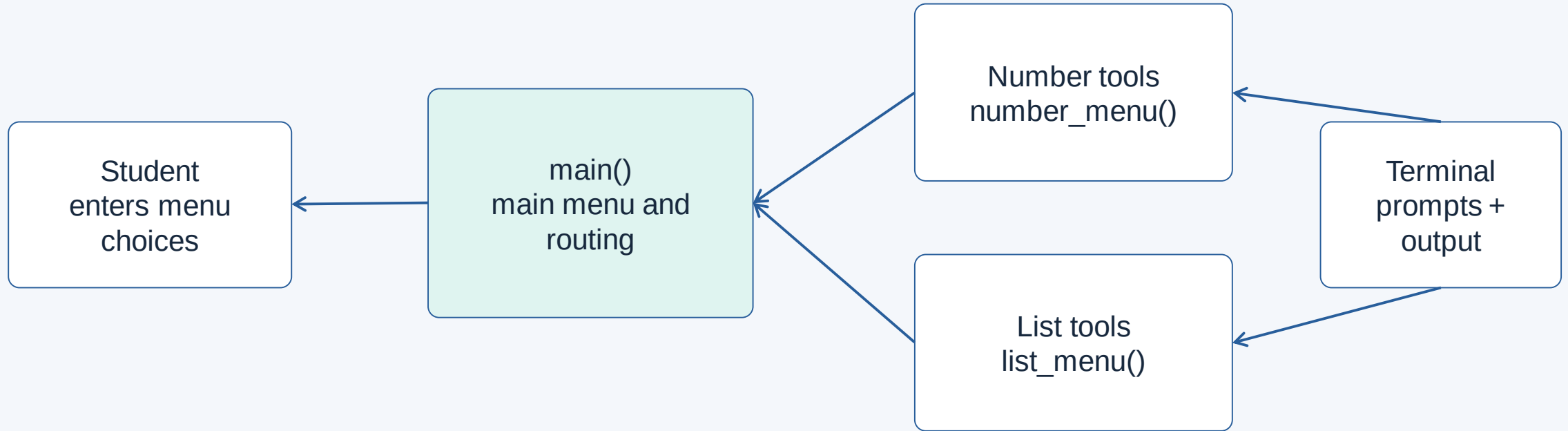
Interaction pattern: choose a tool → enter values → view the result → continue or return to the main menu.

Non-functional requirements

QUALITY	DESIGN EXPECTATION	EVIDENCE / STATUS
Usability	Short menu labels; prompts state the expected input.	Present in CLI flow
Reliability	Catch ValueError and reject selected invalid values.	Partly implemented
Maintainability	Keep operations in named functions and separate menu loops.	Single-file structure
Performance	Use direct built-ins and bounded algorithms for typical small inputs.	No benchmarks supplied
Resource use	Runs locally with Python standard library imports.	No external packages in source

System architecture

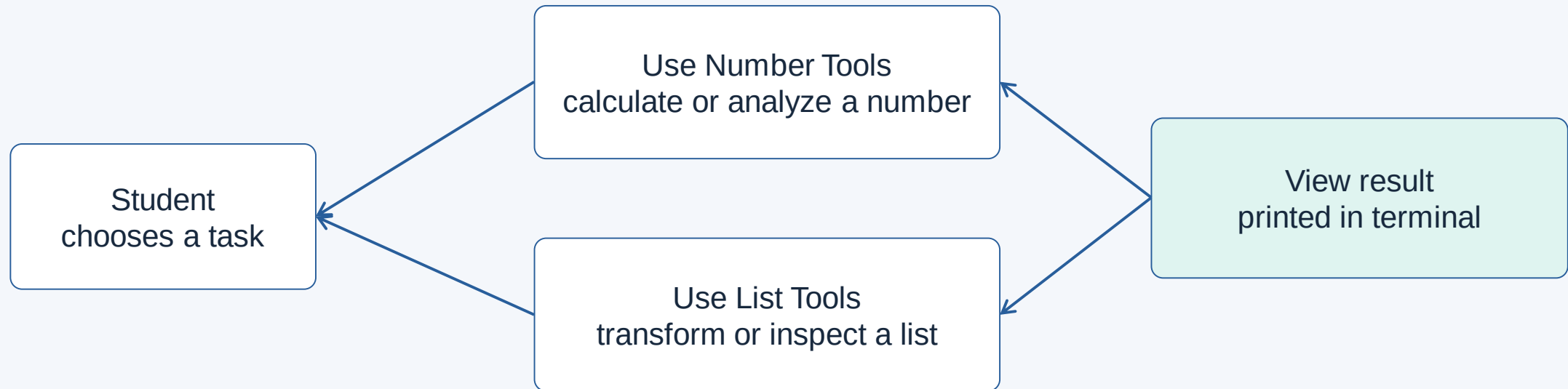
Single-process, local command-line architecture



Execution: one Python process • Storage: none • External services: none

Design diagrams | Use case

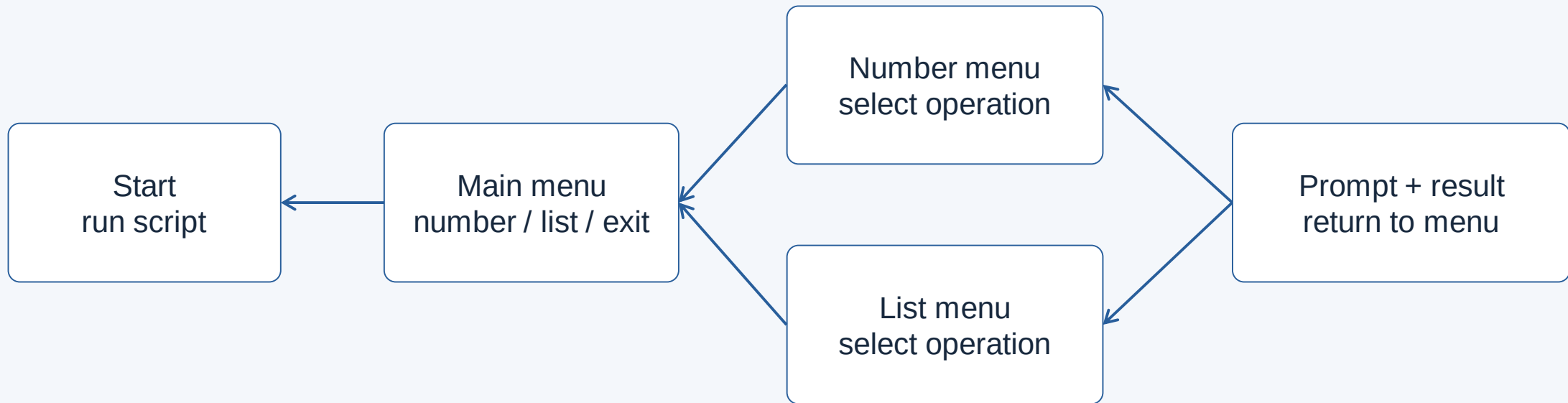
Primary actor: the student operating the terminal



The user provides values through prompts; results are displayed in the same session.

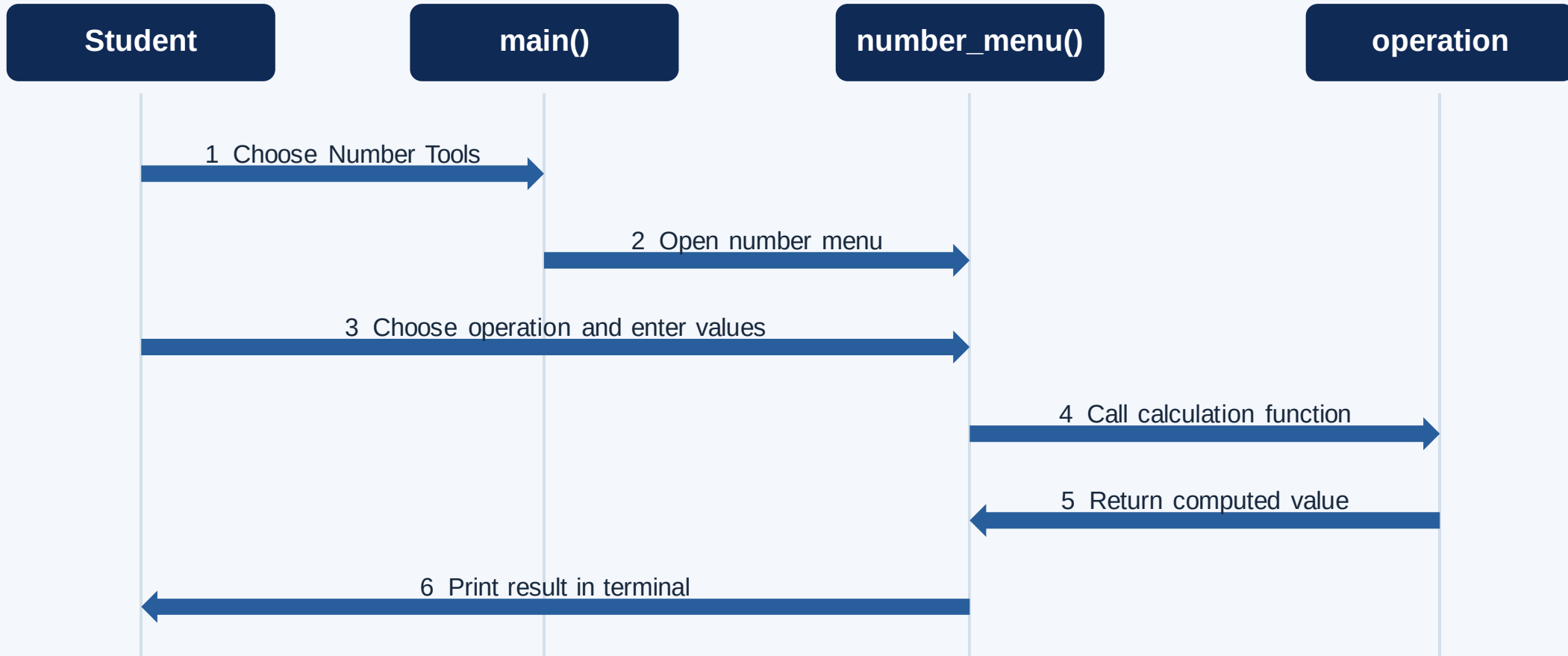
Design diagrams | Workflow

Menu navigation and result loop

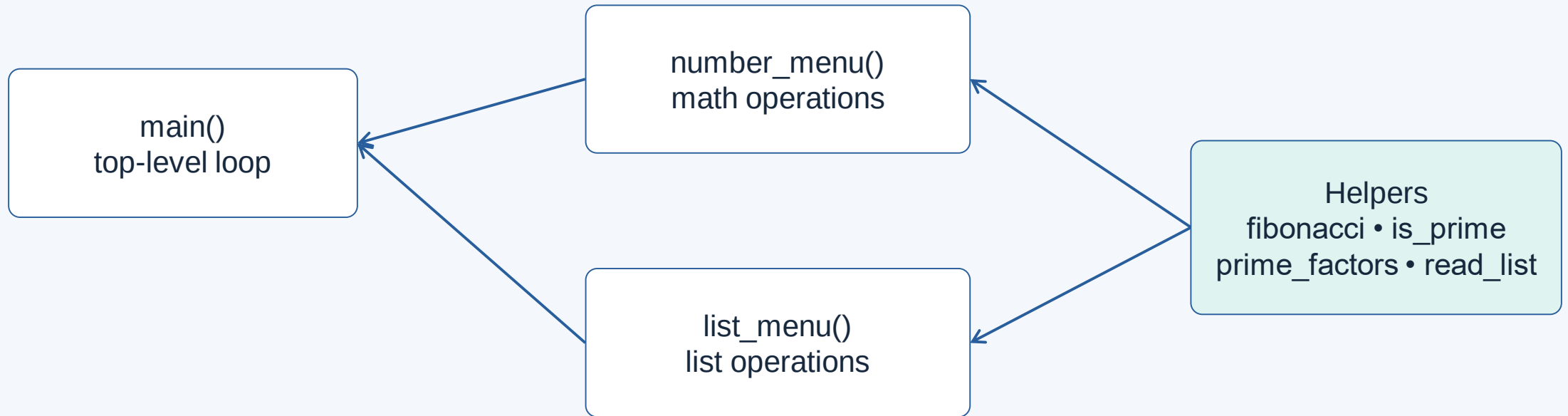


Exit ends the program; choosing “Back” returns to the main menu.

Design diagrams | Sequence



Design diagrams | Components and storage



ER diagram: not applicable — the current program does not store records or use a database.

Design decisions and rationale

Menu-driven CLI

Makes a collection of operations easy to browse without a GUI.

Named functions

Keeps calculation logic and menu flow readable in one source file.

Standard-library math

Uses tested built-ins for GCD, LCM, factorial, and square root.

Python list operations

Uses slicing, sorting, counting, and ordered de-duplication for common tasks.

Try/except ValueError

Keeps selected input and domain errors from immediately terminating the menu.

Implementation details

KEY FUNCTIONS

- `fibonacci(num)`: iterative value; sequence is generated for display
- `is_prime(num)`: trial division through $\sqrt{n}$
- `prime_factors(num)`: repeated division
- `remove_duplicates(values)`: preserves first-seen order
- `read_list()`: parses comma-separated integers

INPUT + EDGE HANDLING

- Integer and float prompts
- Negative input rejected for selected operations
- Empty maximum list and out-of-range `k` are rejected
- `ValueError` messages return control to the menu

Implementation: one Python file, function-based, no external packages or persistent data.

Screenshots and results | Menus

Application launch and available tool groups

```
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code> & C:\Users\HP\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/HP/OneDrive/Desktop/algomate_456.py"

AlgoMate
1. Number tools
2. List tools
0. Exit

Choose: █
```

Main menu | Number Tools • List Tools • Exit

```
Number Tools
1. GCD and LCM
2. Prime check
3. Prime factorization
4. Fibonacci
5. Factorial
6. Square root
7. Power
0. Back

Choose: █
```

Number Tools | seven operations

```
List Tools
1. Reverse list
2. Remove duplicates
3. Find maximum
4. Count occurrences
5. Find kth smallest
6. Sort list
0. Back
```

List Tools | six operations

Screenshots and results | Number tools

Representative outputs from the submitted PowerShell runs

```
Choose: 1  
First number: 45  
Second number: 54  
GCD: 9  
LCM: 270
```

GCD / LCM | inputs 45 and 54 | result: GCD 9, LCM 270

```
Choose: 4  
Fibonacci index: 5  
Value: 5  
Sequence: [0, 1, 1, 2, 3, 5]
```

Fibonacci | index 5 | value 5 and sequence [0, 1, 1, 2, 3, 5]

Screenshots and results | List tools

List transformations and queries

```
Choose: 2  
Enter numbers separated by commas: 1,5,4,6,4,5,2,1  
Without duplicates: [1, 5, 4, 6, 2]
```

Remove duplicates | preserves the first occurrence order in the shown result

```
Choose: 6  
Enter numbers separated by commas: 4,6,5,4,8,5,6,9,4,1  
Sorted list: [1, 4, 4, 4, 5, 5, 6, 6, 8, 9]
```

Sort list | input values are displayed in ascending order

Screenshots and results | Selection and exit

Additional terminal outputs from the submitted run

```
Choose: 5
Enter numbers separated by commas: 45,63,14,2,54,65,36,25,45,67
Enter k: 3
3th smallest value: 25
```

Kth-smallest selection | k = 3 | result: 25

```
AlgoMate
1. Number tools
2. List tools
0. Exit

Choose: 0
Goodbye,hav a great day!
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code>
```

Exit | program prints a goodbye message

Screenshots and results | Other operations

Prime check

121 $\rightarrow$ False; 97 $\rightarrow$ True

Manual run

Prime factorization

3896 $\rightarrow$ [2, 2, 2, 487]

Manual run

Maximum

[45, 6, 5, 48, 41, 2, 14, 51, 421] $\rightarrow$ 421

Manual run

Kth smallest

k = 3 $\rightarrow$ 25

Manual run

Power

5⁴ $\rightarrow$ 625

Manual run

Testing approach

EVIDENCE AVAILABLE

- Manual runs through both top-level menu areas
- Screenshots cover arithmetic, number properties, and list operations
- Observed output examples are included in this report

VALIDATION TO ADD

- Automated unit tests for each helper
- Boundary tests: 0, 1, negatives, empty input, and large values
- Malformed comma-separated input and invalid menu choices
- Regression tests for menu return and exit paths

No automated test files were included in the supplied source. The screenshots document example runs, not a complete test suite.

Challenges faced

Implementation concerns visible in the code

Input parsing

Convert terminal strings into integers or floats and report conversion errors.

Domain boundaries

Handle negative values for factorial, square root, Fibonacci index, and exponent.

Menu continuity

Keep the program available after a result while supporting Back and Exit.

List edge cases

Reject an empty maximum query and invalid kth-smallest positions.

These points are inferred from validation branches in the implementation.

Learnings and key takeaways

The project applies core Python concepts to a practical interactive program.

- Function decomposition and parameter/return-value use
- Loops and conditional routing for nested menus
- Built-in math, list slicing, sorting, counting, and dictionaries
- Exceptions and domain validation for safer interaction
- Manual execution and visible outputs as a first validation step

Future enhancements

- Split the single script into 5–10 focused modules as the project grows
- Add a shared input-validation layer and clearer recovery prompts
- Create unit tests for helpers and menu behavior
- Add documentation: setup, usage examples, and test instructions
- Consider a graphical interface only after the core CLI is tested
- Add optional history or saved datasets if persistent storage is required

The current version is a local CLI and does not persist user data.

References

PROJECT MATERIALS

- AlgoMate source code supplied by Vikram Singh (Python file)
- Student-provided terminal output screenshots, captured 28 Sep 2026
- VITyarthi — Build Your Own Project: General Project Instructions & Submission Guidelines, section 6 and supporting requirements
- Project repository: github.com/sweetvik/algomate

This presentation is organized around all 15 report items listed in the submission brief.